Quality Assurance & Testing Document: Smart Legal Compliance System

1. Introduction

The purpose of this document is to outline the quality assurance (QA) framework and testing standards for the Smart Legal Compliance System. This document ensures that the AI-driven auditing tool meets the required accuracy, performance, and reliability benchmarks for legal compliance against the Malaysia Employment Act.

2. Testing Objectives

Validate Accuracy: Ensure the system maintains an accuracy rate of at least 94.7% in identifying legal violations.

Verify Latency: Confirm that the end-to-end query response time does not exceed 1.2s.

Assess Robustness: Verify the system can process and index over 500+ legal documents without performance degradation.

Ensure Reasoning Integrity: Validate that the AI reasoning chains correctly map retrieved legal context to specific audit findings.

3. Core Test Scenarios

3.1 Data Ingestion & Embedding (Steps 1-2)

Chunking Integrity: Verify that the Malaysia Employment Act documents are split into segments that preserve section context and legal references.

Vector Consistency: Test if text chunks are accurately converted into dense vectors by the embedding models for storage in ChromaDB.

3.2 Retrieval & Semantic Search (Steps 3-5)

Similarity Search Accuracy: Test ChromaDB's ability to retrieve the most relevant legal context based on specific user compliance queries.

Query Vectorization: Ensure the Backend Orchestrator (FastAPI) correctly analyzes and vectorizes various user input formats.

3.3 AI Reasoning & Prompt Engineering (Steps 6-7)

Legal Context Injection: Confirm that relevant provisions of the Malaysia Employment Act are correctly injected into the ILMU AI prompts.

Reasoning Chain Validation: Test the multi-step logical analysis to ensure it moves correctly from identifying regulations to evaluating violations.

Report Structure: Verify that the generated Compliance Reports include necessary citations and actionable recommendations.

4. Performance & Technical Requirements

Test Category	Target Metric	Source Requirement
Response Time	< =1.2 seconds per query	
Audit Accuracy	>=$ 94.7%	
Data Capacity	500+ documents processed	
Reliability	Seamless communication between RAG, ChromaDB, and ILMU AI	

5. Technical Stack Testing

FastAPI Orchestrator: Test API endpoints for consistent output formatting and comprehensive error handling.

ChromaDB: Test high-performance semantic search and embedding storage stability.

ILMU AI: Evaluate the language model's ability to process engineered prompts for complex legal reasoning.

6. Future Expansion Testing

Future QA phases will include regression testing for:

Regional Compliance: Testing the system's adaptability to regional legal datasets.

Multi-Jurisdictional Auditing: Validating the AI's ability to distinguish between different legal jurisdictions during the audit process.